FEAT-ZD-48600 · BE-04 Identify the components/layers to be changed · Components, layers, sources or pipelines to change

Backend agent · item task.components · source hobasa-backend-agent/outputs/FEAT-ZD-48600/task.components.md · approved by automated checks and independent judge

task.components: FEAT-ZD-48600

Components and layers the Backend lane creates or updates for FEAT-ZD-48600 (ZD-48600, "Show which overdue invoices belong to vendors that usually pay late"), with the smallest change that meets the design. This is step BE-04. There is no gate: automated checks, the independent judge and sampled human evaluation review it (record:brief).

Statements marked **agent judgement** are this agent's reading of the code and inputs; no input states them as fact.

Field	Value	Source
Trace ID	FEAT-ZD-48600	record:brief
Step	BE-04 Identify the components/layers to be changed (attempt 1)	record:brief
Inputs	feat.tasks (b0fd4d2), feat.stories (0590ea4), feat.prd (908b58d), feat.trd (3637e51), feat.design (722a98f), task.repo_access (aac590a)	record:brief
Backend tasks	TSK-05 (STORY-05, size M) and TSK-06 (STORY-06, size S), both in ap_aging	feat.tasks (section 1); task.repo_access (section 1)
Repository	ap_aging, Python 3.9 standard library, branch `feature/story` at `5d2cbb9`, working tree clean	task.repo_access (section 2); file:ap_aging/CLAUDE.md
Code read	`src/ap_aging/__init__.py`, `src/ap_aging/aging.py`, `tests/test_aging.py`, `tests/__init__.py`, `ci.py`, `README.md`, `CLAUDE.md`	tool:repo; file:ap_aging/ci.py; file:ap_aging/README.md; file:ap_aging/CLAUDE.md
Result	**Two source files updated, one test file created. Nothing else changes.** No new module, layer, dependency, storage or transport	sections 2 to 4

1. What the existing code is

ap_aging is a single-module library. It has no service or transport layer (feat.design A3 "Transport for CT-1"; feat.trd R-7), no database (task.repo_access section 2), and no configuration file: the repository holds only `src/ap_aging/__init__.py`, `src/ap_aging/aging.py`, `tests/`, `ci.py`, `README.md` and `CLAUDE.md` (tool:repo; file:ap_aging/ci.py; file:ap_aging/README.md).

Layer	File	What it holds today	Source
Package API	`src/ap_aging/__init__.py`	`from .aging import aging_buckets; __version__ = "1.4.0"; __all__ = ["aging_buckets"]`	tool:repo

Layer	File	What it holds today	Source
Domain logic	`src/ap_aging/aging.py`	`BUCKETS`; `days_past_due(due_date, as_of)` = `max((as_of - due_date).days, 0)`; `bucket_for(days)`; `aging_buckets(invoices, as_of)`, which skips `inv.get("paid")` and sums `Decimal(inv["amount"])` per bucket. Imports: `date`, `Decimal`, `Dict`, `Iterable`	tool:repo
Unit tests	`tests/test_aging.py`	`test_days_past_due`, `test_bucket_edges`, `test_aging_buckets_skips_paid`, with `AS_OF = date(2026, 10, 25)`	tool:repo
CI	`ci.py`	Compile, unittest discovery in `tests/`, coverage of `src/ap_aging` with `COVERAGE_MIN = 90.0`, static checks (no `print`, `eval`, `exec`, no bare `except`, lines at most 140 characters) on every `src/ap_aging/*.py`	file:ap_aging/ci.py
Docs	`README.md`	One-line description and the CI command; it lists no functions	file:ap_aging/README.md

Two facts in the code shape the change:

- `bucket_for(30)` returns `"31-60"` because the `"1-30"` branch is `days < 30`. The new overdue rule must not reuse `bucket_for`, or an invoice exactly 30 days past due would be listed (tool:repo; feat.trd F-10, IF-5.3; feat.design DD-4).
- `days_past_due` already gives calendar days after the due date, floored at 0, and `aging_buckets` already uses `inv.get("paid")` for the paid test. Both can be reused as they are (tool:repo; feat.trd IF-5.1, IF-5.2; record:memory D-2026-031, D-2026-050).

2. Components to change

#	Component	Layer	Change	Task / story	Design	Source
CMP-1	`OVERDUE_AFTER_DAYS = 30` in `aging.py`	Domain	**New** module constant	TSK-05 / STORY-05	HC-6; B3.2; DD-4	feat.design (B3.1, B3.2); feat.tasks (TSK-05)
CMP-2	`TWO_PLACES = Decimal("0.01")` in `aging.py`	Domain	**New** module constant, shared by CMP-3 and CMP-4	TSK-05 / STORY-05	B3.2; DD-10	feat.design (B3.2); feat.tasks (TSK-05)
CMP-3	`overdue_with_risk(invoices: Iterable[Dict], as_of: date, risk_by_vendor: Dict[str, str]) -> List[Dict]` in `aging.py`	Domain	**New** pure function	TSK-05 / STORY-05	HC-6; IF-5	feat.design (B3.2); feat.trd (IF-5.1 to IF-5.8); feat.tasks (TSK-05)

#	Component	Layer	Change	Task / story	Design	Source
CMP-4	`overdue_rows_to_json(rows: Iterable[Dict]) -> str` in `aging.py`	Domain (serialisation to CT-1)	**New** pure function	TSK-06 / STORY-06	HC-7; CT-1; DD-3	feat.design (A3, B3.2); feat.trd (IF-6); feat.tasks (TSK-06)
CMP-5	Imports in `aging.py`	Domain	**Updated**: add `import json`; extend the existing `from typing import Dict, Iterable` line with `List`	TSK-05 (`List`), TSK-06 (`json`)	B3.1	feat.design (B3.1); feat.tasks (TSK-05, TSK-06); tool:repo
CMP-6	`src/ap_aging/__init__.py`	Package API	**Updated**: import and add `overdue_with_risk` and `overdue_rows_to_json` to `__all__`; one minor version step from `1.4.0`	TSK-05 (export, version), TSK-06 (export)	B3.1; DD-11	feat.design (B3.1, DD-11); feat.tasks (section 2, TSK-05, TSK-06); task.repo_access (section 3)
CMP-7	`tests/test_overdue.py`	Unit tests	**New** file	TSK-05, TSK-06	B3.5	feat.tasks (section 2, TSK-05, TSK-06); feat.design (B3.5)

Sources: feat.design (A2 HC-6, HC-7; B3.1 to B3.5); feat.tasks (TSK-05, TSK-06); feat.trd (C-6, C-7).

3. Components that do not change

Component	Why it stays	Source
`BUCKETS`, `days_past_due`, `bucket_for`, `aging_buckets`	"Must not change"; existing signatures kept; T-11 regression	feat.tasks (TSK-05); feat.trd (IF-5.8, T-11); file:ap_aging/CLAUDE.md
`tests/test_aging.py`	Existing functions must return the same results as today (AC-05.9), so their tests stay as they are (agent judgement); new tests go to `tests/test_overdue.py` "to limit merge conflicts between parallel tasks"	feat.tasks (section 2); feat.stories (AC-05.9)
`ci.py`	The coverage source (`src/ap_aging`) and static checks already cover the new code; no CI change is needed	file:ap_aging/ci.py
`README.md`	Lists no functions; the new functions carry docstrings instead (agent judgement)	file:ap_aging/README.md
Transport, service or API layer	None exists and none is added; adding one would re-trigger DES-01	feat.design (A3, DD-3, A8); feat.tasks (TSK-06 "Must not change")
Database, storage, configuration, secrets	No repository uses a database; functions are pure and in-memory	task.repo_access (section 2); feat.design (NF-10)
Other repositories (ap_pipeline, ap_forecast, ap_portal)	Other lanes; ap_aging builds against DC-1, DC-3 and CT-1, not their code	feat.design (A3, Part C "Lane dependencies"); task.repo_access (section 1)

4. Minimum-change design per component

The aim is the fewest new lines that meet AC-05.1 to AC-05.10 and AC-06.1 to AC-06.4, reusing what the module already has (agent judgement within feat.design B3.2).

CMP-3 `overdue_with_risk` *(TSK-05)*

Step	Minimal implementation	Reuse / why no extra code	Source
Paid filter	`if inv.get("paid"): continue`	Same test as `aging_buckets`	feat.trd (IF-5.1); tool:repo; record:memory (D-2026-050)
Days	`days = days_past_due(inv["due_date"], as_of)`	Reuses the existing function; no new date arithmetic	feat.trd (IF-5.2); tool:repo; record:memory (D-2026-031)
Overdue rule	`if days > OVERDUE_AFTER_DAYS`	Strict > excludes exactly 30; never `bucket_for`	feat.trd (IF-5.3); feat.design (DD-4); feat.stories (AC-05.2)
Row	`{"id", "vendor", "days", "amount": Decimal(inv["amount"]).quantize(TWO_PLACES), "risk": risk_by_vendor.get(inv["vendor"], "unknown")}`	`Decimal(...)` as in `aging_buckets`; no float	feat.trd (IF-5.4, IF-5.5); feat.design (B3.2); record:memory (D-2026-044, L-011)
Missing `vendor`	No explicit check: `inv["vendor"]` raises `KeyError` by itself	No added branch, so no extra line to cover (agent judgement)	feat.trd (IF-5.7); feat.stories (AC-05.10); feat.tasks (TSK-05)
Sort	`rows.sort(key=lambda row: (-row["days"], row["id"]))`	One sort; no secondary pass	feat.trd (IF-5.6); feat.design (B3.2, NF-11)
`as_of`	Required parameter, no default	—	feat.tasks (TSK-05)

CMP-4 `overdue_rows_to_json` *(TSK-06)*

One `json.dumps` over a list comprehension that builds each object in CT-1 key order: `id`, `vendor`, `days` as `int(r["days"])`, `amount` as `str(Decimal(r["amount"]).quantize(TWO_PLACES))`, `risk`. An empty list gives `'[]'` with no special case. No date field, no float (feat.design B3.2, CT-1; feat.trd IF-6; feat.stories AC-06.1 to AC-06.4; feat.tasks TSK-06).

CMP-6 `__init__.py`

Extend the existing import line to `from .aging import aging_buckets, overdue_rows_to_json, overdue_with_risk`, add both names to `__all__`, and step `__version__` one minor version from `1.4.0`; the exact next number is for the lane to confirm (feat.design B3.1, DD-11; feat.tasks section 2; tool:repo).

CMP-7 `tests/test_overdue.py`

Tests named in feat.tasks TSK-05 and TSK-06, with `AS_OF = date(2026, 10, 25)` and due dates built as `AS_OF - timedelta(days=N)` so that `days` is 30, 31, 42 and 75 (feat.tasks TSK-05; feat.design B3.5):

Test	Covers	Source
`test_overdue_boundary_30_and_31`	AC-05.1, AC-05.2; L-007	feat.tasks (TSK-05); feat.design (B3.5); record:memory (L-007)
`test_overdue_excludes_paid_and_not_due`	AC-05.3, AC-05.4	feat.tasks (TSK-05); feat.design (B3.5)
`test_overdue_row_keys_and_sort`	AC-05.5, AC-05.6 (tie on 42: `SYN-INV-002` before `SYN-INV-003`)	feat.tasks (TSK-05); feat.design (B3.5)
`test_overdue_unknown_risk`	AC-05.7	feat.tasks (TSK-05); feat.design (B3.5)
`test_overdue_amount_quantised`	AC-05.8 (`Decimal("980.5")` → `Decimal("980.50")`)	feat.tasks (TSK-05); feat.design (B3.5)

Test	Covers	Source
`test_overdue_missing_vendor_raises`	AC-05.10	feat.tasks (TSK-05); feat.design (B3.5)
`test_existing_aging_unchanged`	AC-05.9; pins `bucket_for(30) == "31-60"`, which `tests/test_aging.py` does not test today	feat.tasks (TSK-05); feat.design (B3.5); tool:repo
`test_rows_to_json`	AC-06.1 to AC-06.3 (`"1250.00"` for SYN-INV-004, `days` an integer, `[]` for no rows, key order); compare `json.loads` values	feat.tasks (TSK-06); feat.design (B3.5)

Every line of CMP-3 and CMP-4 is reached by these tests, so the new code does not lower the package coverage below the `COVERAGE_MIN = 90.0` gate (agent judgement; the CI run in the coding steps confirms it) (file:ap_aging/ci.py; feat.tasks section 2).

5. Change map

```
flowchart LR
  subgraph API["__init__.py (package API)"]
    I1["__all__ + overdue_with_risk, overdue_rows_to_json<br/>__version__ minor step"]
  end
  subgraph DOM["aging.py (domain)"]
    E1["days_past_due (unchanged, reused)"]
    E2["bucket_for / aging_buckets / BUCKETS (unchanged, not used)"]
    N0["OVERDUE_AFTER_DAYS, TWO_PLACES (new)"]
    N1["overdue_with_risk (new)"]
    N2["overdue_rows_to_json (new)"]
    E1 --> N1
    N0 --> N1
    N0 --> N2
    N1 -- rows --> N2
  end
  N1 --> I1
  N2 --> I1
  T1["tests/test_overdue.py (new)"] -.-> N1
  T1 -.-> N2
  T1 -.-> E2
```

6. Build order and commits

Order	Task	Files	Commit prefix	Source
1	TSK-05	`aging.py` (CMP-1, CMP-2, CMP-3, `List` import), `__init__.py` (export, version), `tests/test_overdue.py`	`STORY-05`	feat.tasks (TSK-05, section 2); task.repo_access (section 3)
2	TSK-06	`aging.py` (CMP-4, `json` import), `__init__.py` (export), `tests/test_overdue.py`	`STORY-06`	feat.tasks (TSK-06, section 2); feat.stories (section 2)

Both on `feature/story`, as the interim answer to RAQ-1 (task.repo_access section 5). CI bar for each: compile, tests, coverage of at least 90%, static checks, Python 3.9 syntax only (feat.tasks section 2; file:ap_aging/ci.py).

7. Prerequisites and stop conditions

Item	Status	Source
Repository and write access	Met	task.repo_access (section 4)
Database change	None needed	task.repo_access (section 2)
Dependency on other lanes	None in code; DC-1 and DC-3 are plain dicts	feat.design (A3, Part C); feat.tasks (TSK-05)

Item	Status	Source
Design or AC ambiguity that stops coding	None. Pending items (PQ-2, TQ-3, TQ-9) have interim answers in the design	feat.stories (section 3); feat.design (D3)

8. Differences between inputs, resolved

Point	Inputs	Followed	Source
Test file	feat.stories TK-05.2 proposes `tests/test_aging.py` (agent judgement; SA-6 allows a new file); feat.tasks assigns new `tests/test_overdue.py`	`tests/test_overdue.py` (later, task-level input)	feat.stories (TK-05.2, SA-6); feat.tasks (section 2)
JSON amount expression	feat.stories TK-06.1: `str(amount.quantize(Decimal("0.01")))`; feat.design: `str(Decimal(r["amount"]).quantize(TWO_PLACES))`	feat.design form; same result for a `Decimal` row, and it also accepts a hand-built row	feat.stories (TK-06.1); feat.design (B3.2); feat.tasks (TSK-06)

Neither changes behaviour, so neither is a conflict with an approved output.

9. Open questions (non-blocking)

ID	Question	Interim answer	Owner (proposed)
CQB-1	Exact next version number for ap_aging	One minor step from `1.4.0`, proposed in the coding step for the lane to confirm	Developer
CQB-2	Carried: TQ-3 (CT-1 approval and serialisation owner), TQ-9 (`KeyError` for missing `vendor`), PQ-2 (sort tie-break), DQ-2 (rounding mode), RAQ-1 (one branch for both stories)	Interim answers as in feat.design D3, feat.trd 7.2 and task.repo_access section 5	Architect / PM / Developer

No blocking questions.